Impact of a real-time prescription benefit on adherence and utilization of low-cost prescription alternatives for members new to diabetes treatment

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Plain language summary

This study looked at whether a tool called real-time prescription benefit (RTPB) helps people with diabetes. RTPB shows the cost of medications to doctors when they prescribe. The study found that using RTPB helped people take their medicine more, use more generic (cheaper) drugs, and spend less money. Both patients and health plans save money when doctors used RTPB.

Implications for managed care pharmacy

This study shows that RTPB tools can improve medication adherence, increase the use of cost-effective generic drugs, and reduce both patient and plan costs. These findings can inform benefit design by promoting RTPB integration into prescribing systems. RTPB may also support formulary development by guiding prescribers to choose affordable clinically appropriate alternatives, potentially reducing health disparities by improving access to lower-cost medications.

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ABSTRACT

BACKGROUND: Chronic diseases such as diabetes are a major burden to the US health care system. High medication adherence helps improve diabetes outcomes and reduce cost. Cost of medications can contribute to nonadherence. Use of a formulary decision support system with e-prescribing may be associated with greater use of generic medications, leading to lower costs and better adherence. A real-time prescription benefit (RTPB) solution provides patient-specific drug pricing, benefit information, and therapeutic options to choose the most cost-effective and clinically appropriate treatment.

OBJECTIVES: To examine whether RTPB is associated with increased adherence measured by proportion of days covered, higher utilization of generics, and generic dispensing rate? Is RTPB associated with lower plan and patient out-of-pocket (OOP) per-user per-month costs?

METHODS: This study used a retrospective, matched intervention-control analysis of commercial health plan members from a large pharmacy benefits manager. Members were eligible for inclusion if they initiated therapy between January and August 2021. Members were excluded if they were not continuously eligible for coverage over the study period. Members who initiated diabetes therapy with a prescriber using RTPB (intervention) were compared with those new to therapy with a prescriber not using RTPB (control). Index date for both samples was the first medication prescription in the index period. Members were matched on age and sex demographics. The evaluation period lasted 12 months after index date. Multivariable linear regression models were used to assess the impact of an RTPB program on adherence and proportion of prescriptions filled with a generic. A generalized linear model (gamma distribution, log link) estimated plan and OOP patient costs, whereas a generalized linear model model with the Poisson distribution was used to estimate the number of controlling for patient age, sex, social determinants of health score, and other patient- and plan-level covariates.

RESULTS: 1,302 matched pairs were included in the analysis. Findings show the proportion of days covered was 68.7% for control and 71.4%

ABSTRACT *continued*

for RTPB members (P<0.05). The average number of generic prescriptions for control and RTPB samples were 4.06 and 5.66, respectively (P<0.05) and the generic dispensing rates were 44.9% and 60.1%, respectively (P<0.05). The mean plan cost per member per month for diabetes medications, for the non-RTPB group, was 32.3% higher than the RTPB sample (a difference of $81.69, P<0.0001) and the mean patient cost per month was 88.8% higher than the RTPB sample (a difference of $9.71, P<0.0001).

CONCLUSIONS: Access to RTPB tools provides prescribers with formulary benefit and therapeutic options that allow them to provide the lowest-cost clinical treatment, thus improving adherence, increasing use of generic medications, and lowering plan and patient OOP costs.

Chronic diseases like diabetes are among the leading causes of death in the United States.[1] Approximately 16% of adults older than 20 years have diabetes (diagnosed and undiagnosed). The burden of diabetes on our health care system is high, with economical costs topping $300 billion dollars in 2017 and accounting for more than 38.2 million office visits in 2019.[1,2] A study of 20,326 National Health Interview Survey participants with diabetes found that 17.6% (2.3 million) of those younger than 65 years and 6.9% (0.7 million) of those 65 years and older reported cost-related medication nonadherence: about 1 in 6 younger adults and 1 in 14 older adults in the United States. Addressing financial barriers to medication access could improve adherence, leading to better health outcomes and lower costs. Effective diabetes management is essential for both improving patient outcomes and reducing health care costs, with medication adherence playing a key role.[3–5] Better adherence not only lessens the disease burden but also enhances health outcomes and lowers costs for both patients and payers.[6–8]

Many patients cite that rising and unknown drug costs affect their adherence, which can lead to prescription abandonment.[9–11] Interventions such as mail-order pharmacies or prescribing an affordable and appropriate alternative can be effective in addressing cost-related barriers.[12–14] Tools, such as a formulary decision support system, at the point of prescribing have shown better outcomes, including cost savings, increased adherence, and better therapeutic results.[15–17] A real-time prescription benefit (RTPB) is a formulary decision support system solution, embedded in the electronic health record system, that shows prescribers, at the point of prescribing via an e-prescribing platform (eg, Surescripts, ArriveHealth, CenterX), patient-specific drug pricing, benefit information, and therapeutic options to assist in choosing the most appropriate and clinically beneficial treatment. Use of RTPB is associated with lower patient out-of-pocket (OOP) costs and with increased rates of prescription pick-up.[18,19] For patients with chronic conditions like diabetes, medication adherence is necessary for optimal clinical outcomes. Most newer agents and pharmacotherapeutic options on the market are more costly than previous agents but with better cardiovascular benefits; RTPB may be an effective resource to achieve therapeutic targets, as it can address patient financial concerns. This study aims to examine whether the RTPB solution directed toward the prescriber will increase formulary compliance, use of the most beneficial channel (either retail or home delivery pharmacy), and increase medication adherence for patients new to diabetes pharmacotherapy.

Methods

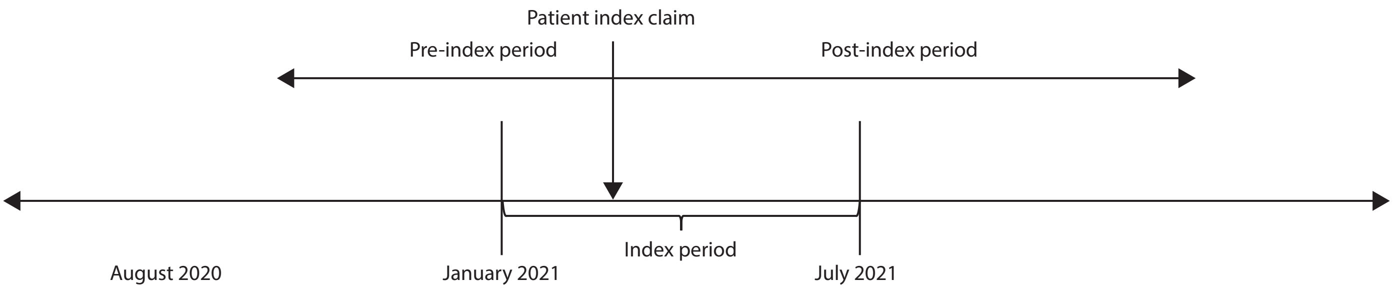
DESIGN, SETTING, AND PARTICIPANTS

This study used a retrospective, intervention/control design to investigate the impact of RTPB tools on medication use and costs among individuals newly initiating diabetes therapy. The study population consisted of members enrolled in a Pennsylvania commercial health plan with pharmacy benefits continuously managed by a large pharmacy benefits manager between 2020 and 2022. Members were classified into 2 groups: the intervention group, comprising individuals whose prescribers used RTPB tools (from 1 of 3 platforms: Surescripts, Arrive Health, or CenterX) during the index claim dispensing, and the control group, comprising individuals whose prescribers did not use RTPB tools. Utilization was tracked through transaction data from the electronic health record, which was linked to an e-prescribing platform and connected to claims data.

The index date for both groups was defined as the date of the first diabetes medication prescription filled between January 1 and July 31, 2021 (Figure 1). To be included in the study, members were required to have no diabetes prescriptions filled in the 6 months preceding the index date (ie, preperiod), indicating new initiation of diabetes therapy. The evaluation period extended for 12 months following the index date (ie, postperiod), allowing for assessment of medication utilization and cost outcomes over a full year.

To ensure comparability between the intervention and control groups, members were matched on a 1:1 basis based on demographics, using the greedy algorithm on age and sex of the sample participants, as well as indicators for preperiod utilization of total and generic medications, defined by the number of adjusted (30-day) prescriptions.

FIGURE 1 Study Timeline



Exclusion criteria for the study included members not continuously eligible for pharmacy benefits coverage throughout the study period. This ensured that medication utilization and cost data were consistently captured for all participants, minimizing potential biases due to variations in benefit coverage.

INTERVENTION

The primary exposure of interest in this study was the utilization of RTPB tools by prescribers. Utilization was defined as a prescriber using the RTPB system to review information on lower-cost formulary or channel alternatives; however, it did not necessarily indicate that the prescriber selected an alternative. RTPB tools provide real-time information on medication formulary coverage, OOP costs, and potential therapeutic alternatives during the prescribing process. By comparing outcomes between patients whose prescribers used RTPB and those whose prescribers did not, the study aimed to assess the impact of this technology on generic medication selection, adherence, and associated costs.

OUTCOMES

The study evaluated the effect of RTPB on 3 key outcomes: (1) total costs incurred by the health plan and patients for diabetes medications, termed plan and patient OOP costs; (2) utilization of generic diabetes medications; and (3) medication adherence.

Plan and patient OOP costs were calculated as the total amount paid by the health plan and patients, respectively, for all filled diabetes medications during the evaluation period. Utilization of generic diabetes medications was assessed using 2 metrics: (1) the mean number of generic diabetes prescriptions filled per member during the evaluation period and (2) the generic dispensing rate (GDR), defined as the proportion of total diabetes medication prescriptions that were dispensed as generics. Medication adherence was measured using the proportion of days covered (PDC), which represents the percentage of days within the evaluation period for which the patient had medication coverage, calculated based on the number of days covered and total days in the evaluation period. To ensure the validity of PDC calculations, criteria were established requiring members to have either 2 or more 30-day adjusted prescriptions (including the index claim) or a minimum of 64 days supply of medication within the first 90 days of the evaluation period.

ANALYSIS

To evaluate the relationship between RTPB program utilization and study outcomes, multivariable linear regression models were used to assess the impact of an RTPB program on adherence and proportion of prescriptions filled with a generic. A generalized linear model (gamma distribution, log link) estimated plan and OOP patient costs, whereas a GLM model with a Poisson distribution was used to estimate the number of controlling for patient age, sex, social determinants of health (SDOH) score, and other patient- and plan-level covariates.

These models allowed for the assessment of the independent effect of RTPB tools on plan and patient OOP costs, utilization of generic diabetes medications, and medication adherence while controlling for the influence of potential confounding factors. Covariates included in the models encompassed demographics (age and sex), SDOH index, enrollment in diabetes care management programs (care management programs optimize formulary placement of diabetes medications, implement utilization management, and promote cost-effective, convenient channels like home delivery), formulary type (Administrative Services Only, Fully Insured, Affordable Care Act on and off the Exchange, Medicare and Medicaid), and variables to control for the number of adjusted prescriptions for the different types of diabetes medications; biguanides, DPP4 inhibitors, incretin mimetic drugs, sodium-glucose cotransporter-2 (SGLT2) inhibitors, sulfonylureas, and thiazides. The SDOH indicator is a composite score ranging from 1 to 100, with higher scores signifying a greater burden of social factors negatively impacting health outcomes (Supplementary Table 1, available in online article).[20] To facilitate comparisons across medications with varying durations of action, all prescription counts were standardized to 30-day equivalents. SAS version 9.4 was used for data processing and analyses (SAS Institute Inc.).

Results

STUDY POPULATION

The final study cohort comprised 1,302 matched pairs of members who initiated diabetes therapy during the index period and met all inclusion and exclusion criteria (Supplementary Table 1). The demographic characteristics of the RTPB and non-RTPB groups were well balanced, with a mean age of 53.4 years in both groups and an equal proportion of male participants (43.0%). However, a statistically significant difference was observed in the SDOH indicator, with the RTPB group exhibiting a slightly higher mean score as compared with the non-RTPB group, respectively (53.25 vs 52.37, $P = 0.0017$).

Further analysis of baseline characteristics revealed notable differences in enrollment in care management programs and health plan types between the two groups. A larger proportion of members in the non-RTPB group (55.2%) were enrolled in a diabetes care management program compared with the RTPB group (38.8%), a difference that was statistically significant ($P < 0.0001$). Conversely, the RTPB group had higher proportions of members enrolled in Medicare Part D plans (22.8% vs 15.3%, $P < 0.0001$) and Medicaid plans (34.7% vs 27.6%, $P < 0.0001$) compared with the non-RTPB group, respectively.

OUTCOME FINDINGS

Bivariate Plan and Patient OOP Costs. An analysis of medication costs revealed substantial differences between RTPB and non-RTPB groups. The mean plan cost per month was 32.3% higher in the non-RTPB compared with the RTPB group (a difference of $81.69, $P < 0.0001$), whereas the mean patient cost per month was 88.8% higher than the RTPB sample (a difference of $9.71, $P < 0.0001$).

Utilization of Generic Diabetes Medications. Regarding generic medication utilization, significant differences were observed between the two groups. The RTPB group filled a mean of 5.66 generic prescriptions during the evaluation period, compared with a mean of 4.06 generic prescriptions in the non-RTPB group (Supplementary Figure 1) ($P < 0.0001$). Furthermore, the GDR, calculated as the percentage of diabetes medication prescriptions dispensed as generics, was higher in the RTPB group compared with the non-RTPB group, respectively, 60.1% vs 44.9%, $P < 0.0001$) (Supplementary Figure 2). The average number of adjusted prescriptions by diabetes drug class varied between the RTPB and control samples (9.3 vs 10.1, respectively), with biguanides being the most used (3.8 vs 3.5), followed by incretin mimetics (2.3 vs 2.7) and SGLT2s (1.1 vs 2.0).

Medication Adherence. The outcome measure of medication adherence, measured as the PDC, was significantly higher in the RTPB group compared with the non-RTPB group ($P < 0.0001$) (Supplementary Figure 3). The mean PDC for members in the RTPB group was 71.4%, whereas the mean PDC for members in the non-RTPB group was 68.7%.

MULTIVARIABLE MODEL FINDINGS

In the multivariable linear regression and GLM analyses, after adjusting for potential confounding factors, several significant associations between RTPB program utilization and study outcomes were observed. In terms of costs, results of GLM models indicated that members whose prescribers used the RTPB program experienced a significant reduction in both mean monthly member OOP costs and mean monthly plan costs (Supplementary Table 2).

Furthermore, RTPB utilization was associated with a substantial increase in the use of generic medications. The mean number of generic prescriptions filled per user per month was 1.6 higher in the RTPB group compared with the non-RTPB group ($P < 0.0001$) (Supplementary Figure 1). Additionally, the mean percentage of claims filled with generic medications was 9.3% higher in the RTPB group ($P < 0.0001$) (Supplementary Figure 2).

Finally, medication adherence, as measured by the PDC, was also significantly improved with the RTPB program. Unadjusted adherence was 2.7% higher in the RTPB sample ($P < 0.0001$) (Figure 1). Although multivariable linear regression results showed members in the RTPB group had an adjusted adherence that was 3.4% higher, on average, PDC compared to those in the non-RTPB group ($P \leq 0.0001$), controlling for all study covariates (Supplementary Table 2).

Discussion

This study examined the impact of RTPB tools on medication use and costs among individuals newly initiating diabetes therapy. The findings suggest that RTPB utilization is associated with several significant benefits for both patients and health plans.

The most notable finding was the significant reduction in OOP costs for both patients and plans among members whose prescribers used RTPB tools. This is consistent with previous research suggesting that RTPB tools can lead to lower medication costs by facilitating the selection of more cost-effective alternatives at the point of prescribing.[11,18,21] The cost savings observed in this study are consistent with what has been reported in some prior studies, highlighting the potential financial impact of widespread RTPB adoption.

Additionally, the study found that RTPB utilization was associated with a significant increase in the use of generic medications, both in terms of the average number of generic prescriptions filled and the overall GDR. This finding is particularly important given the potential for generic medications to reduce health care spending without compromising therapeutic effectiveness.[22] The increased generic utilization observed in the RTPB group suggests that these tools may help prescribers identify and select generic alternatives that align with both clinical guidelines and patient preferences. This finding is particularly relevant considering the baseline differences in insurance coverage observed between the two groups, with the non-RTPB group having a higher proportion of members enrolled in Medicare Part D and Medicaid plans. These plans often have different formulary structures and cost-sharing arrangements compared with commercial plans, which could influence medication choices and utilization patterns.[23-25]

Furthermore, the study demonstrated a significant improvement in medication adherence among patients whose prescribers used RTPB tools. This finding is consistent with the hypothesis that providing real-time information on medication costs and coverage may empower patients to make more informed decisions about their treatment, leading to greater adherence.[11] Improved medication adherence is a critical factor in achieving optimal clinical outcomes for patients with chronic conditions such as diabetes.[4-8] The observed difference in baseline enrollment in diabetes care management programs between the two groups, with higher enrollment in the RTPB group, warrants further investigation to understand the potential interplay between RTPB tools and care management programs in influencing medication adherence and clinical outcomes.[18,26]

LIMITATIONS

Several limitations should be noted. First, the retrospective observational design of the study limits causal inference, and it is possible that unmeasured confounding factors may have contributed to the observed associations. For instance, prescribers who choose to use RTPB tools may be more likely to engage in other behaviors that promote medication adherence and cost savings, such as providing detailed medication counseling or actively seeking out generic alternatives for their patients. These unmeasured provider behaviors could act as confounders, making it difficult to isolate the independent effect of the RTPB program itself. Additionally, although the study controlled for a variety of potential confounders, including demographics and an SDOH index, the possibility of residual confounding remains. Although the index claims in the RTPB sample were selected based on the physician requesting a list of potential savings options, it was unclear whether they ultimately chose a lower-cost alternative. Utilization was defined as a prescriber accessing the RTPB system to explore lower-cost formulary or channel options, but this did not necessarily indicate that they opted for one. Additionally, diabetes diagnosis was obtained through prescription use only, as there was no access to medical claims or diagnoses for these patients. Although the SDOH index aimed to capture the influence of social factors on health outcomes, it may not have fully accounted for all relevant variables, potentially leading to residual confounding. Finally, it is not known what the impact of RTPB tools on long-term patient outcomes will be. For example, greater medication compliance could be associated with long-term improved clinical outcomes. Conversely, if RTPB tools led to substitution of an evidence-based, expensive branded product for an alternative not known to improve clinical outcomes, long-term clinical outcomes could decline. Therefore, future research using randomization of larger, more diverse study populations would be valuable in confirming and extending these findings.

Conclusions

Despite the limitations inherent in a retrospective study design, this study highlights the potential benefits of integrating RTPB tools into clinical practice, suggesting that RTPB tool use by prescribers can lead to significant benefits for both patients and health plans. The observed reductions in medication costs, increased use of generics, and improved medication adherence underscore the potential value of incorporating an RTPB program into routine clinical practice. Continued research and implementation efforts are needed to better understand the impact of RTPB technology on health care delivery, affordability, quality, and patient outcomes. Future studies should examine whether RTPB tool use increases the likelihood of patients being prescribed preferred formulary medications, such as SGLT2 inhibitor or glucagon-like peptide-1 receptor agonists, because their higher costs could lead to use of lower-cost alternatives that might not be appropriate.

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